EE2002-Tutorial 2 Solution Key

1. Given a pair of terms for which a variable appears in one term and the complement of that variable in another, the *consensus* term is formed by multiplying the two original terms together, leaving out the selected variable and its complement. For example, the consensus of *ab* and *a'c* is *bc*. We can show that the consensus term is redundant in a Boolean expression. That is, a Boolean expression of the form XY + X'Z + YZ is equivalent to XY + X'Z. Show this via a truth table.

Solution:

We will demonstrate the equivalence of XY + X'Z + YZ and XY + X'Z via a truth table.

X	Y	Z	XY	$X'Z$	YZ	$XY + X'Z$	$XY + X'Z + YZ$
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	1
0	1	0	0	0	0	0	0
0	1	1	0	1	1	1	1
1	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0
1	1	0	1	0	0	1	1
1	1	1	1	0	1	1	1

Alternate method

We can write XY + X'Z + YZ as (XY + X'Z + (X + X')YZ) (since X+X'=1)

Step 1: Expand $(X + X')YZ$

$$(X + X')YZ = XYZ + X'YZ$$

Substituting back:

$$XY + X'Z + XYZ + X'YZ$$

Step 2: Group terms

$$(XY + XYZ) + (X'Z + X'YZ)$$

Step 3: Factor common terms

$$XY(1 + Z) + X'Z(1 + Y)$$

Step 4: Simplify using Boolean Identities

$$1 + Z = 1, \quad 1 + Y = 1$$

Thus we have

$$XY + X'Z$$

2. Show that the following equation (in Boolean algebra) is valid

$A'BC'D+(A'+BC)(A+C'D')+BC'D+A'BC' = ABCD+A'C'D'+ABD+ABCD'+BC'D$

Solution:

Left-hand side:

$$A'BC'D + (A' + BC)(A + C'D') + BC'D + A'BC'$$

This simplifies to:

$$ABC + A'C'D' + BC'D$$

Right-hand side:

$$ABCD + A'C'D' + ABD + ABCD' + BC'D$$

Also simplifies to

$$ABC + A'C'D' + BC'D$$

3. Simplify $(A'B(D' + C'D) + B(A + A'CD))$ to one literal.

Solution: This simplifies to B

4. Simplify Boolean expression $(A' + C)(A' + C')(A + B + C'D)$.

Solution: This simplifies to A'(B + C'D)

5.Given the Boolean function $F = xy'z + x'y'z + w'xy + wx'y + wxy$:

(a) Obtain the truth table of the function.

(b) Draw the logic diagram for the Boolean expression.

(c) Simplify the function using Boolean Algebra.

(d) Draw the logic diagram for the simplified form.

(e) Compare the total number of gates (inverters, OR gates, and AND gates) for the two cases: the original Boolean function and the simplified form.

Solution

(a) Truth Table for Each Term

We construct the truth table for each individual term in F, write the logic terms of $F = 1$ that are $(1, 5, 6, 7, 9, 10, 11, 13, 14, 15)$:

w	x	y	z	$xy'z$	$x'y'z$	$w'xy$	$wx'y$	wxy	F
0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	1	0	0	0	1
0	0	1	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	1
0	1	0	1	1	0	0	0	0	1
0	1	1	0	0	0	1	0	0	1
0	1	1	1	0	0	1	0	0	1
1	0	0	0	0	0	0	0	0	1
1	0	0	1	0	1	0	0	0	1
1	0	1	0	1	0	0	1	0	1
1	0	1	1	0	0	0	1	0	1
1	1	0	0	0	0	0	0	0	1
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1	1	1	1	0	0	0	0	1	1

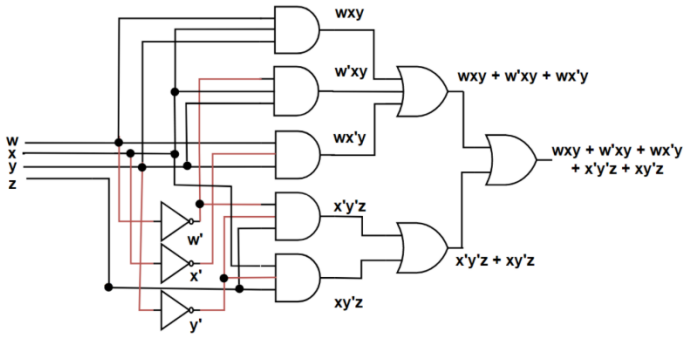


Figure 1: Non-Simplified Boolean Function Representation

(c) Simplification of F

This can be simplified to:

$$F = xy + wy + y'z.$$

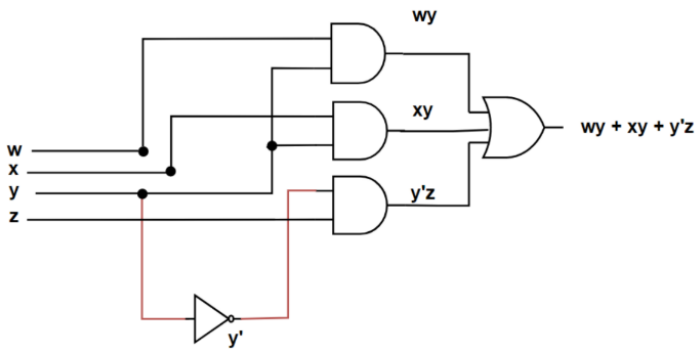


Figure 2: Simplified Boolean Function Representation

(e) Gate Count Comparison:

Gate Type	Original Expression	Simplified Expression
3-input AND	5	0
2-input AND	0	3
3-input OR	1	1
2-input OR	2	0
NOT	3	1
Total Gates	11	5

6.Find the complement of the following expressions: (a) $xy' + x'y$ and (b) $(AB' + C)D' + E$.

Solution (a): The complement is:

$$(x' + y)(x + y').$$

Solution (b): The complement is:

$$[(A' + B)C' + D]E'.$$

7. Consider the following Boolean function expressed as the sum of minterms.

$$F = A'BC + AB'C + AC'B'$$

Express this Boolean function as the product of maxterms.

Solution:

$$F = (A + B + C)(A + B + C')(A + B' + C)(A' + B' + C)(A' + B' + C').$$

8.Obtain the minterm expansion of the following Boolean function using a truth table. Write the solution using the $\sum$ notation. Also obtain the maxterm expansion and write the same using the $\prod$ notation.

$$\mathbf{F = X'(Y + Z') + XY'}$$

Solution: The minterm expansion is written as:

$$F = \sum m(0, 2, 3, 4, 5).$$

Step 3: Maxterm Expansion: The maxterm expansion is written as:

$$F = \prod M(1, 6, 7).$$

9.Prove DeMorgan's Laws using a Truth Table

DeMorgan's laws state the following:

1. $(A \cdot B)' = A' + B'$
2. $(A + B)' = A' \cdot B'$

(a) Proof of $(A \cdot B)' = A' + B'$
The truth table for $(A \cdot B)' = A' + B'$ is shown below:

A	B	$A \cdot B$	$(A \cdot B)'$	A'	B'	$A' + B'$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

From the truth table, it can be observed that the columns for $(A \cdot B)'$ and $A' + B'$ are identical.

(b) Proof of $(A + B)' = A' \cdot B'$
The truth table for $(A + B)' = A' \cdot B'$ is shown below:

A	B	$A + B$	$(A + B)'$	A'	B'	$A' \cdot B'$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

From the truth table, it is evident that the columns for $(A+B)'$ and $A' \cdot B'$ are identical.

10.The output of a majority circuit is 1 if a majority (more than half) of its inputs are equal to 1. The output is 0 otherwise. Construct a truth table for a three-input majority circuit and derive a simplified sum-of-products expression for its output.

Solution

Assuming the inputs are A, B and C and the output is F, we have

$$F = BC + AB + AC$$